

2 Homotopy of $\mathcal{C}_p(G)$

In this section, we review some of the standard facts of order complexes of posets. See the first two parts of Smith's book[9] for more details.

Let $(\mathcal{P}, \leq)$ be a poset (short for partially ordered set). The order complex $\Delta\mathcal{P}$ of $\mathcal{P}$ consists of all finite chains in $\mathcal{P}$, including the empty set. The chain of length i in $\mathcal{P}$ consisting of $i + 1$ elements is called an i -simplex in $\Delta\mathcal{P}$. In particular, the empty set $\emptyset$ is the (-1) -simplex. It is well-known that such (abstract) simplicial complex $\Delta\mathcal{P}$ can be embedded into a Euclidean topological space as a subspace denoted by $|\Delta\mathcal{P}|$, so-called the geometric realization of $\Delta\mathcal{P}$. We abbreviate $|\Delta\mathcal{P}|$ as $|\mathcal{P}|$ here.

A poset map $f : (\mathcal{P}, \leq_{\mathcal{P}}) \rightarrow (\mathcal{Q}, \leq_{\mathcal{Q}})$ of posets $\mathcal{P}, \mathcal{Q}$, is a set-theoretical map such that $x \leq_{\mathcal{P}} y$ implies $f(x) \leq_{\mathcal{Q}} f(y)$. It is routine to check that a poset map $f : \mathcal{P} \rightarrow \mathcal{Q}$ naturally induces a simplicial map Δf between order complexes $\Delta\mathcal{P}$ and $\Delta\mathcal{Q}$, and f also induces a continuous map $|f| : |\mathcal{P}| \rightarrow |\mathcal{Q}|$. The following lemma gives a sufficient condition for the homotopy between two continuous maps induced by poset maps.

Lemma 2. [9, Lemma 3.1.7] *Suppose that $f, g : P \rightarrow Q$ satisfy f, g are poset maps with $f \leq g$, that is, $f(x) \leq g(x)$ for each $x \in P$. Then f and g are homotopic, that is, $|f|, |g| : |\mathcal{P}| \rightarrow |\mathcal{Q}|$ are homotopic.*

Recall that topological spaces X and Y are homotopy equivalent if there are continuous maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ whose compositions satisfy the homotopies $g \circ f \simeq \text{Id}_X$ and $f \circ g \simeq \text{Id}_Y$. For posets $\mathcal{P}, \mathcal{Q}$, $\mathcal{P}$ and $\mathcal{Q}$ are said to be homotopy-equivalent if $|\mathcal{P}|$ and $|\mathcal{Q}|$ are homotopy-equivalent, simply denoted by $\mathcal{P} \simeq \mathcal{Q}$.

For a prime p and a group G , we denote

$$\mathcal{I}_p(G) = \{P_1 \cap P_2 \cdots \cap P_s \mid P_i \in \text{Syl}_p(G) \text{ for all } 1 \leq i \leq s, \text{ and } s \geq 1\}$$

by the set of all intersections of some Sylow p -subgroups of G .

Lemma 3. *Let G be a group and p be a prime. Write*

$$\mathcal{C}(\mathcal{I}_p(G)) = \{Hx \mid H \in \mathcal{I}_p(G), x \in G\}.$$

Then $\mathcal{C}_p(G)$ is homotopy-equivalent to $\mathcal{C}(\mathcal{I}_p(G))$.

Proof. For each $Hx \in \mathcal{C}_p(G)$, we define a map f from $\mathcal{C}_p(G)$ to $\mathcal{C}(\mathcal{I}_p(G))$ by

$$f(Hx) := \bigcap_{H \subseteq P \in \text{Syl}_p(G)} Px = \left(\bigcap_{H \subseteq P \in \text{Syl}_p(G)} P \right) x.$$